

Assignment 1

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Problem 1

First let's define the two random variables, and their respective probability distribution. For the random variable U , the number of tries before the first success can be expressed from a geometric distribution which takes two parameters: k the number of trials, and p the probability of success. Thus,

$$U \sim \text{Geometric}(k, p) \quad (1)$$

For the random variable V , the number of trials needed to get 2 heads can be expressed as a negative binomial distribution which takes three parameters: r the number of success, x the number of failure, and p the probability of success per trial. Thus

$$V \sim \text{NBinom}(r, x, p) \quad (2)$$

From the definition of the problem, we know that $S = \{\text{head}, \text{tail}\}$, and

$$P(\omega) = \begin{cases} \frac{1}{2} & \text{if } \omega = \text{head} \\ \frac{1}{2} & \text{if } \omega = \text{tail} \end{cases} \quad (3)$$

where p is the probability of obtaining head.

```
# Set the seed
set.seed(48109)
# Number of simulations
sims <- 1000
# Number of trials
trials <- 100
# Probability of success
p <- 0.5
# Define empty vector for U
u <- c()
```

```

# Define empty vector for V
v <- c()

# Initiate the loop
for (i in 1:sims) {
  heads <- 0
  trials <- 0
  first_head <- NA

  while (heads < 2) {
    trials <- trials + 1
    flip <- rbinom(1, 1, p)
    if (flip == 1) {
      heads <- heads + 1
      if (heads == 1) first_head <- trials
    }
  }
  u[i] <- first_head
  v[i] <- trials
}

# Plotting the output
plot(
  jitter(u) ~ jitter(v),
  xlab = "Number of trials to get 2 heads",
  ylab = "Number of trials before 1st head"
)
abline(lm(u ~ v), col = "red")

# Testing the correlation
cor(u, v, method = "spearman")

```

```
[1] 0.7178861
```

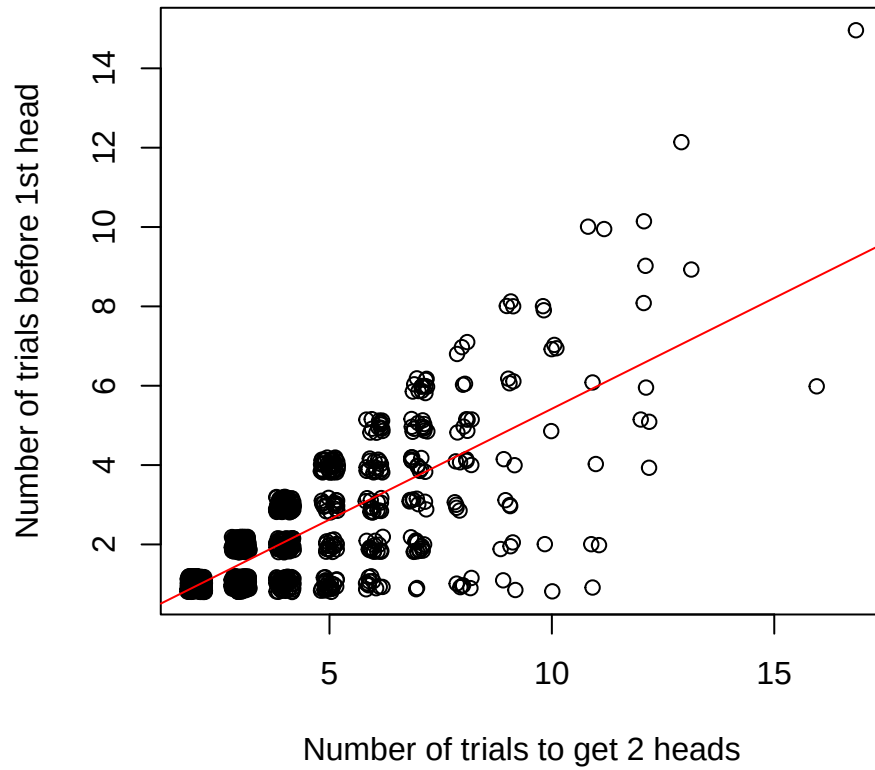


Figure 1: Joint distribution of the random variables U and V

From Figure 1, we can see that the two random variables are positively correlated. The Spearman correlation also confirm this intuition as the correlation is 0.71.

Problem 2

Part A